

rameters used for scaling and shifting learned by gDS can generalize to unseen classes. During training, we randomly select half of the categories from IMAGENET-INSTAGRAM-mini and use them to train the gDS. At test time, we evaluate the learned model on the other half of categories that have never been seen before from IMAGENET-INSTAGRAM validation set, and on all categories as well. We compare with a pretrained ResNet50 and ResNet50-ft under the same settings and the results are shown in Table. 4.

We can see that on unseen categories, the performance of ResNet50 finetuned using filtered images on seen categories decreases, while gDS increases the performance by 4.24% compared to ResNet50. This demonstrates that gDS not only improves the generalization of CNNs on filtered images but also across different categories. When tested on all categories, finetuned ResNet50 improves upon ResNet50 slightly, but still worse than gDS.

Method	Other Half	All
	Top1/Top5 acc	Top1/Top5 acc
ResNet50	67.64/ 87.85	67.25/ 87.67
ResNet50 ft	59.96/ 84.67	68.89/ 89.10
gDS	71.40/ 90.49	71.08/ 90.30

Table 4: Performance of models learned with only half of the object categories, and evaluated on the other half of the object categories as well as all categories.

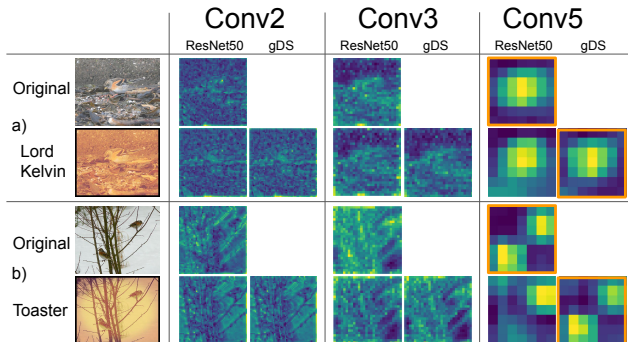


Figure 6: Feature maps comparison of ResNet50 and ResNet50+gDS. Two cases from two most challenging Instagram filters are shown. (a) “Lord Kelvin”, (b) “Toaster”. For each example case, we show the original IMAGENET image and its Instagram-filtered version. For each image, we also show the activation feature maps after Conv2, Conv3 and Conv5, generated by ResNet50 and Resnet50+gDS. gDS successfully transforms the features of filtered images to be close to original images (See the diagonal comparisons).

Comparisons with alternative methods. We compare gDS with two alternative methods based on ResNet50, *i.e.*, AdaBN (?) and AdaIN (?) that perform alignment without retraining the weights of the whole network. AdaBN (?) accumulates BN layer statistics on the target domain, without

accessing training samples from IMAGENET-INSTAGRAM. However, since AdaBN is designed for a single target domain, we apply AdaBN on the validation set of each filter type separately. In this way, AdaBN targets each Instagram filter type at a time, thus obtaining better performance than applying AdaBN on entire IMAGENET-INSTAGRAM. On the other hand, AdaIN performs style transfer on filtered images such that the styles generated by filters are removed by an image generator. More specifically, we randomly select 100 images per Instagram filter from IMAGENET-INSTAGRAM training set and 200 images from IMAGENET to train a generator. Then the generator is used to synthesize filter-free images, upon which the original pretrained CNN is applied. The results are shown in Table. 5.

Method	Top1/Top5 acc
ResNet50	67.21/ 87.62
AdaBN (?)	68.87/ 88.74
AdaIN (?)	35.45/ 58.86
gDS	71.41/ 90.46

Table 5: Comparison with other alternative methods on IMAGENET-INSTAGRAM-mini.

We can see AdaBN improves the performance of pre-trained models by only 1% where as gDS obtained a 4.2% performance gain. The results of AdaIN are worse than directly applying ResNet50. A reason could be that the synthesis process with an image generator is far from perfect and further introduces artifacts and distribution shifts.

Conclusion

We presented a study on how popular filters that are prevalent on social media affect the performance of pretrained modern CNN models. We created IMAGENET-INSTAGRAM, by applying 20 pre-defined ImageNet filters to each image in IMAGENET. We found that filters induce significant differences in feature maps compared to those of original images, and further lead to significant drops when directly applying CNNs pretrained on IMAGENET. To improve generalization, we introduced a lightweight de-stylization module, which produces parameters used to scale and shift feature maps in order to recover changes brought by filters. Combining the lightweight module together with skip connections, we presented gDS that can be plugged into modern CNN networks. Extensive studies are conducted on IMAGENET-INSTAGRAM and the results confirm the effectiveness of the proposed method.

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Quisquam unde qui praesentium sunt, ea at commodi possimus dignissimos minima laboriosam cum, asperiores veritatis aliquid